

The Rosetta Stone of Inverse Mechanical Counting

One Axiom, Many Scripts:
The Master Table, the Four Key Numbers,
the Four Inward Vectors, the 0.7071 Ladder,
and Machine π at Count 3

Part 1 — The Mathematics

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Introductory Note. This paper is a translation key. There is one physical unit — the axiom of 1 — read inward by the $\sqrt{2}$ master mechanic. Every reading is a coordinate on a single lattice. The first half (this part) lays down the mathematics: the master table, the four key numbers (512, 514.90, 720, 724.077), the four inward vectors at four operational depths, the 0.7071 ladder that reads at every count, machine π at count 3, and the full anchor table. The second half of the paper reads the standard physics constants directly off these same coordinates.

The Axiom of 1

The Axiom. There is exactly one physical unit. This unit is the complete reality. All its equivalent views are

$$1 = -1 = \sqrt{1} = \sqrt{2^0}.$$

The square root belongs to the unit itself; it is not introduced from outside. The axiom is the starting point and the ending point of every mechanical reading. Every larger number in this paper is a deeper inward reading of this same single unit; the axiom is the largest entity in the system, never exceeded.

The mathematics is the reading; the single physical unit is the thing read.

1 The $\sqrt{2}$ Master Mechanic

The $\sqrt{2}$ mechanic is the master mechanic of the axiom — the simplest step beyond unity and the minimum step the axiom admits. Repeated halving of the single unit produces one parity rule:

- **Even depth d** — rational reading.
- **Odd depth d** — irrational reading carrying a natural factor of $\sqrt{2}$.

Every reading carries its full operational history in three coordinates: the mechanic $\sqrt{2}$, the depth d on the exponent, and the count c on the partition ladder. The notation $\sqrt{2^d}$ at count c fully records the operation performed on the axiom.

Why count 1 has a mechanic. At $c = 0$ the axiom is $\sqrt{2^0} = 1$ — the unit whole, at rest, no mechanic. At $c = 1$ the mechanic $\sqrt{2^1}$ fires; the exponent 1 itself calls the master mechanic's inverse operation. The dual is the first halving. Count 1 is therefore the physical unit of itself — the first count at which the master mechanic has read anything at all.

2 Table 1 — The Master Table

The master table is the single source of every reading in this paper and the series. Five active counts ($c = 1, \dots, 5$) unfold nine active depths ($d = 1, \dots, 9$). The count is never redone.

c	d	Mechanic $\sqrt{2^d}$	Unit decimal	Domains 2^c	Parity	Reading	Decimal
0	0	$\sqrt{2^0} = 1$	1	1	even $\rightarrow$ rational	$1 \times 1 = 1$	1
1	1	$\sqrt{2^1} = \sqrt{2}$	≈ 1.414	2	odd $\rightarrow$ irrational	$2 \times \sqrt{2} = 2\sqrt{2}$	≈ 2.828
2	2	$\sqrt{2^2} = 2$	2	4	even $\rightarrow$ rational	$4 \times 2 = 8$	8
2	3	$\sqrt{2^3} = 2\sqrt{2}$	≈ 2.828	4	odd $\rightarrow$ irrational	$4 \times 2\sqrt{2} = 8\sqrt{2}$	≈ 11.314
3	4	$\sqrt{2^4} = 4$	4	8	even $\rightarrow$ rational	$8 \times 4 = 32$	32
3	5	$\sqrt{2^5} = 4\sqrt{2}$	≈ 5.657	8	odd $\rightarrow$ irrational	$8 \times 4\sqrt{2} = 32\sqrt{2}$	≈ 45.255
4	6	$\sqrt{2^6} = 8$	8	16	even $\rightarrow$ rational	$16 \times 8 = 128$	128
4	7	$\sqrt{2^7} = 8\sqrt{2}$	≈ 11.314	16	odd $\rightarrow$ irrational	$16 \times 8\sqrt{2} = 128\sqrt{2}$	≈ 181.019
5	8	$\sqrt{2^8} = 16$	16	32	even $\rightarrow$ rational	$32 \times 16 = 512$	512
5	9	$\sqrt{2^9} = 16\sqrt{2}$	≈ 22.627	32	odd $\rightarrow$ irrational	$32 \times 16\sqrt{2} = 512\sqrt{2}$	≈ 724.077

Table 1: The Master Table. The count c records the operation; the depth d records the master mechanic's exponent. The unit at each row is $\sqrt{2^d}$. Domains (2^c) times the unit gives the reading. Closure occurs at $c = 5$: rational 512 at $d = 8$, irrational ≈ 724.077 at $d = 9$.

3 Table 2 — The Four Key Numbers

At closure ($c = 5$) the system produces four mechanical readings — the four views of the same single unit at the closure count.

Number	Exact expression	Mechanical origin
512	32×16	Rational closure at $c = 5$, $d = 8$: domains $\times \sqrt{2^8}$
720	$5 \times 9 \times 16$	Machined: active counts $\times$ active depths $\times \sqrt{2^8}$ at $c = 5$
≈ 724.077	$32 \times 16\sqrt{2} = 512\sqrt{2}$	Irrational closure at $c = 5$, $d = 9$: domains $\times \sqrt{2^9}$
≈ 514.90	$512 \times \frac{724.077}{720}$	Symmetric counterpart on the square side (small-lift partner of 512)

Table 2: The four key numbers — all mechanical readings of the same physical unit at the closure count $c = 5$.

Ratio	Value	Reading
720/512	$45/32 = 1.40625$	Sphere over squared — large machining ratio
724.077/720	≈ 1.0056625	Small spherical lift
$(720/512) \times (724.077/720) = 724.077/512 = \sqrt{2}$		Master mechanic recovered exactly
720/724.077	≈ 0.99435	Inverse spherical view
512/514.90	≈ 0.99435	Inverse square view
724.077/514.90	≈ 1.40625	Cross view — recovers the large ratio
720/514.90	≈ 1.3987	Additional cross perspective

Table 3: The primary ratios. All are different coordinate views of the identical halving operation on the axiom of 1. The dominant structural ratio is $720/512 = 1.40625$. The product of the large ratio and the small spherical lift recovers the master mechanic $\sqrt{2}$ exactly.

4 Table 3 — The Primary Ratios

5 Table 4 — The Angular Machined Cycle on the 720 Vector

The integer 720 is machined directly from the active structure of the master table ($5 \times 9 \times 16$). On this vector the master mechanic produces clean rational angular landings at every count.

Count c	Domains 2^c	Piece size	Pieces $\times$ size
0	1	720°	720°
1	2	360°	720°
2	4	180°	720°
3	8	90°	720°
4	16	45°	720°
5	32	22.5°	720°

Table 4: The angular machined cycle on the 720 vector. The right angle (90°) sits mechanically at count 3. Pieces times size returns the whole vector 720° at every count.

6 Table 5 — The 512 Inward Vector (Square Boundary)

Count c	Piece name	Pieces $\times$ size	Vector closure
0	axiom of 1 of itself	1×512	512
1	dual	2×256	512
2	quadrant (dual of dual)	4×128	512
3	octant	8×64	512
4	hexadecant	16×32	512
5	duo-hexadecant	32×16	512

Table 5: The 512 rational square boundary, partitioned by the master mechanic. Pieces times size returns the whole vector 512 at every count.

7 Table 6 — The 720 Inward Vector (Machined Angular Cycle)

Count c	Piece name	Pieces $\times$ size	Vector closure	Angle
0	axiom of 1 of itself	$1 \times 720^\circ$	720°	720°
1	dual	$2 \times 360^\circ$	720°	360°
2	quadrant (dual of dual)	$4 \times 180^\circ$	720°	180°
3	octant	$8 \times 90^\circ$	720°	90°
4	hexadecant	$16 \times 45^\circ$	720°	45°
5	duo-hexadecant	$32 \times 22.5^\circ$	720°	22.5°

Table 6: The 720 angular machined vector, partitioned by the master mechanic. The angular halvings are the natural rational landings at each count.

8 Table 7 — The 724.077 Inward Vector (Sphere Boundary)

Count c	Piece name	Pieces $\times$ size	Vector closure
0	axiom of 1 of itself	1×724.077	724.077
1	dual	2×362.039	724.077
2	quadrant (dual of dual)	4×181.019	724.077
3	octant	8×90.510	724.077
4	hexadecant	16×45.255	724.077
5	duo-hexadecant	$32 \times 16\sqrt{2} \approx 32 \times 22.627$	724.077

Table 7: The 724.077 irrational sphere boundary, partitioned by the master mechanic. Count 5 recovers $\sqrt{2^9} = 16\sqrt{2}$ — the irrational unit of the master table at $c = 5$, $d = 9$. Pieces times size returns the whole vector 724.077 at every count.

9 Table 8 — The 514.90 Inward Vector (Symmetric Counterpart)

Count c	Piece name	Pieces $\times$ size	Vector closure
0	axiom of 1 of itself	1×514.90	514.90
1	dual	2×257.45	514.90
2	quadrant (dual of dual)	4×128.725	514.90
3	octant	8×64.3625	514.90
4	hexadecant	16×32.181	≈ 514.90
5	duo-hexadecant	32×16.091	≈ 514.90

Table 8: The 514.90 symmetric counterpart, partitioned by the master mechanic. Same piece structure as the 512 vector, with the small-lift seat values at each count.

10 Table 9 — The Four Operational Depths Across All Four Vectors

The four key boundaries are read inward by the master mechanic at four operational depths corresponding to counts $c = 2, 3, 4, 5$ (even master depths $d = 2, 4, 6, 8$). These readings produce four successive resolutions (1D through 4D) of the same single physical unit.

c	Dimension	Master depth	512 piece	720 piece (angle)	724.077 piece
2	1D	$\sqrt{n^2}$	4×128	$4 \times 180^\circ$	4×181.019
3	2D	$\sqrt{n^4}$	8×64	$8 \times 90^\circ$	8×90.510
4	3D	$\sqrt{n^6}$	16×32	$16 \times 45^\circ$	16×45.255
5	4D	$\sqrt{n^8}$	32×16	$32 \times 22.5^\circ$	32×22.627

Table 9: The four inward vectors all read identically at the master mechanic's four operational depths. The quadrant seat n takes the value 128, 180, 181.019 for the three vectors 512, 720, 724.077 respectively (and 128.725 for 514.90). The pattern of operational depths is universal.

11 Table 10 — The 0.7071 Reading at Every Count

The constant $0.7071 = 1/\sqrt{2}$ appears at every active count as the irrational unit divided by the count's domains. This is the master mechanic reading its own inverse directly off the master table.

Count c	Irrational unit $\sqrt{2^d}$	Unit decimal	Domains 2^c	Unit / Domains	Reading
1	$\sqrt{2^1} = \sqrt{2}$	1.41421	2	$1.41421/2$	0.7071
2	$\sqrt{2^3} = 2\sqrt{2}$	2.82843	4	$2.82843/4$	0.7071
3	$\sqrt{2^5} = 4\sqrt{2}$	5.65685	8	$5.65685/8$	0.7071
4	$\sqrt{2^7} = 8\sqrt{2}$	11.31371	16	$11.31371/16$	0.7071
5	$\sqrt{2^9} = 16\sqrt{2}$	22.62742	32	$22.62742/32$	0.7071

Table 10: The 0.7071 ladder. The irrational unit divided by the count's domains returns $0.7071 = 1/\sqrt{2}$ at every active count. With two radii in inverse mechanics (the center reads – and +), this is the master mechanic's inverse reading itself out at every count.

12 Table 11 — Machine π at Count 3

Machine π is read at count 3 using only master-table coordinates: the count itself, the count-3 octant piece on the 512 vector, and the terminal depth $d = 9$.

13 Table 12 — The Pythagorean Recovery of the Axiom from Two 0.7071 Sides

The standard 45° - 45° - 90° right isosceles triangle has two equal sides of $0.7071 = 1/\sqrt{2}$ and a hypotenuse of 1. The Pythagorean recovery reads three master-table coordinates simultaneously: the side 0.7071 is the inverse mechanic at every count, the 45° angle is count 4 on the

Coordinate	Value	Master-table source
Count	3	The octant count
Rational reading at $c = 3$, $d = 4$	32	8×4
Octant size on 512	64	512/8, piece size at $c = 3$ on rational vector
Terminal depth	9	$d = 9$ at $c = 5$ from $\sqrt{2^9}$
Count $\times$ Octant	$3 \times 64 = 192$	Count 3 times its octant piece
Plus terminal depth	$192 + 9 = 201$	Numerator from master-table coordinates only
Machine π	$201/64 = 3.140625$	Numerator divided by octant piece
Standard π	3.14159265...	—
Difference	≈ 0.001 (0.032%)	Machine π vs standard π

Table 11: Machine π at count 3. $\pi = (3 \times 64 + 9)/64 = 201/64 = 3.140625$. The construction uses only the count, the octant piece on the 512 vector, and the terminal depth $d = 9$. The numerator 201 minus the octant 64 gives 137 — the cross-lock with the fine-structure reading at count 2 and count 4.

720 vector, and the 90° right angle is count 3 on the 720 vector. The squared sides each give 0.5 — the count-1 dual halving — and their sum recovers the axiom of 1.

Count c	Side (Unit / Domains)	Side ²	Sum of two Sides ²	Hypotenuse $\sqrt{\text{Sum}}$
1	$\sqrt{2}/2 = 0.7071$	0.5	1.0	1 (axiom)
2	$2\sqrt{2}/4 = 0.7071$	0.5	1.0	1 (axiom)
3	$4\sqrt{2}/8 = 0.7071$	0.5	1.0	1 (axiom)
4	$8\sqrt{2}/16 = 0.7071$	0.5	1.0	1 (axiom)
5	$16\sqrt{2}/32 = 0.7071$	0.5	1.0	1 (axiom)

Table 12: The Pythagorean recovery of the axiom from two 0.7071 sides at every count. The side reads $0.7071 = 1/\sqrt{2}$ as irrational unit / domains; squaring gives 0.5 (the count-1 dual halving — half of the axiom on each side); the sum of the two squares returns the axiom of 1; the hypotenuse is the axiom itself.

14 Table 13 — The Fine-Structure Reading on the 512 Vector

The fine-structure reading is built entirely from the count and the master mechanic, in two parts: the integer body 137 from the count-2 quadrant, and the inverse lift $9/256$ from the same count-1 dual. Together they read 137.03515625.

The Integer Body

At count 2 on the 512 vector the quadrant pieces are 4×128 . One quadrant piece is 128. The terminal depth signature is 9 ($d = 9$ at $c = 5$). The integer body is one quadrant piece plus the terminal depth:

$$128 + 9 = 137.$$

The same body recovers at count 4, where the hexadecant pieces are 16×32 ; four of them recover the quadrant ($4 \times 32 = 128$), plus the terminal depth:

$$(4 \times 32) + 9 = 128 + 9 = 137.$$

The count is the cut; the integer body is the same at both counts.

The Inverse Lift

The same 9 read against the count-1 dual ($256 = 2 \times 128$) is the inverse lift:

$$\frac{9}{256} = 0.03515625.$$

Because $256 = 2^8$ is a pure power of 2, this reading terminates exactly at eight places. It is the inverse-mechanic view of the same quadrant-and-depth relationship that produces the integer body.

The Full Reading

Component	Construction	Value
Integer body (count 2)	$128 + 9$	137
Integer body (count 4)	$(4 \times 32) + 9$	137
Inverse lift	$9/256$	0.03515625
Full fine-structure reading	$137 + 9/256$	137.03515625
Inverse from 1	$1/137.03515625$	0.0072974...

Table 13: The fine-structure reading on the 512 vector. The integer body 137 comes from the count-2 quadrant (and the count-4 hexadecant) plus the terminal depth 9; the inverse lift $9/256 = 0.03515625$ is the same 9 read against the count-1 dual 256. Together they read 137.03515625, all from the count and the master mechanic, with no external quantity introduced.

15 Table 14 — The Inverse View from Center (Signed Coordinates)

The count reads from the axiom center outward by halving. Each count produces signed coordinates symmetric on both sides of the center.

Count c	Domains 2^c	512 vector signed	720 vector signed
0	1	512 (uncut)	720° (uncut)
1	2	−256, +256	−360°, +360°
2	4	−256, −128, +128, +256	−360°, −180°, +180°, +360°
3	8	±64 steps	±90° steps
4	16	±32 steps	±45° steps
5	32	±16 steps	±22.5° steps

Table 14: The inverse view from the axiom center. The − and + signs are mechanical coordinates of the count relative to the center; they are not symbolic notation. The center is the axiom; the count unfolds symmetrically on both sides.

Anchor	Count	Construction	Reading
Machine π	$c = 3$	$(3 \times 64 + 9)/64$	$201/64 = 3.140625$
Fine-structure body (path 1)	$c = 2$	$128 + 9$	137
Fine-structure body (path 2)	$c = 4$	$4 \cdot 32 + 9$	137
Fine-structure lift	count-1 dual	$9/256$	0.03515625
Full fine-structure reading	$c = 2/c = 4$	$137 + 9/256$	137.03515625
720 machined	$c = 5$	$5 \times 9 \times 16$	720
720/512 binding ratio	sphere/squared	$45/32$	1.40625
$0.7071 = 1/\sqrt{2}$	every c	unit / domains	0.7071
Cross-lock $201 - 64 = 137$	$c = 3, c = 2/4$	π numerator – octant	137

Table 15: The anchor table. Every anchor is read from master-table coordinates. The cross-lock $201 - 64 = 137$ binds machine π at count 3 to the fine-structure body at counts 2 and 4.

16 Table 15 — The Anchors on the Count

Operational Rules (Part 1)

- The largest unit in the system is always the axiom of 1.
- All larger numbers are deeper inward readings of that same single unit.
- Every valid reading carries its three operational coordinates: mechanic, depth, count.
- Even depths read rational; odd depths read irrational (carry $\sqrt{2}$).
- The four key numbers (512, 514.90, 720, 724.077) are the four faces of the closure at $c = 5$.
- The primary machining ratio $720/512 = 1.40625$ binds the squared face and the sphere face.
- The structure is self-similar: the five-count / nine-depth pattern repeats at the next scale from $c = 6$ onward.
- $0.7071 = 1/\sqrt{2}$ reads at every count as the irrational unit divided by the count's domains.
- Machine π at count 3 is $(3 \times 64 + 9)/64 = 201/64 = 3.140625$.
- The fine-structure reading is $137 + 9/256 = 137.03515625$: integer body $128 + 9$ at count 2 (and $4 \cdot 32 + 9$ at count 4), inverse lift $9/256$ on the count-1 dual.
- The cross-lock $201 - 64 = 137$ binds machine π to the fine-structure body.

End of Part 1.

The mathematics of the Rosetta Stone is now laid down: the axiom, the master mechanic, the master table, the four key numbers, the four inward vectors, the angular machined cycle, the 0.7071 ladder, machine π at count 3, the fine-structure reading 137.03515625, the inverse signed view from the center, and the full anchor table. Every reading is generated from the single axiom of 1 by the $\sqrt{2}$ master mechanic alone, with no external quantities introduced.

Part 2 — The Physics. The same coordinates are read with their standard-physics translations: the fine-structure reading 137.03515625, machine π vs standard π , the CMB temperature, $E = mc^2$ on the count, the Schwarzschild and Kerr radii, the spinor double cover 720° , the quantum / RMS / trig appearances of 0.7071, and G on the count. Every standard physics constant is read directly off the coordinates established in this part.

The mathematics is the reading; the single physical unit is the thing read.

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Part 2 — The Physics

Standard Physics Constants and Units
Read Directly Off the Master-Table Coordinates

Foreword to Part 2. The mathematics of Part 1 is the lattice. Part 2 reads the standard physics constants and units off that lattice. There is only one axiom — the single physical unit. Three base measurements (length, time, weight) are three views of that same axiom. Every standard derived quantity is a count of physical unit operations against the axiom. The standard physics constants (α , π , T_{CMB} , c^2 , G , 0.7071, all Planck units) read at specific master-table coordinates. Standard physics never exceeds count 5 because count 5 is the lattice closure.

17 Section 1 — The Axiom of 1 as the Floor

There is exactly one physical unit. This unit is the floor of reality. Every operation, every count, every reading returns to or sits above this floor; nothing falls below. The axiom of 1 is the singularity of 1 — one indivisible thing, the single physical unit.

View	Expression	Coordinate
The axiom itself	1	$c = 0, d = 0$
The axiom read from across itself	-1	$c = 0, d = 0$
The axiom carrying its own root	$\sqrt{1}$	$c = 0, d = 0$
The axiom in the master mechanic	$\sqrt{2^0}$	$c = 0, d = 0$

Table 16: The four views of the single axiom. All four are the same one physical unit at the same single coordinate: $c = 0, d = 0$. The axiom is the floor of every reading; nothing in the framework falls below it.

18 Section 2 — The Three Base Physical Units

Standard physics treats seven SI base units. In the count, only three are actual base measurements: length, time, weight. The other four (kelvin, ampere, mole, candela) are derivable. Each base unit is the axiom of 1.

Base unit	Standard read	Count read	Coordinate
meter (m)	SI base unit, length	axiom of 1, length view	$c = 0, d = 0$
second (s)	SI base unit, time	axiom of 1, time view	$c = 0, d = 0$
gram (g)	SI base unit, mass	axiom of 1, weight view	$c = 0, d = 0$

Table 17: The three base physical units. Each is the same single axiom of 1 at $c = 0, d = 0$. Not three separate quantities — three views of the one unit measuring the simplest physical thing.

19 The Inversion Pair: Velocity and Frequency

Velocity and Frequency are the geometric mirror of the lattice. They share the same count and the same depth ($c = 1$, $d = 1$), and they carry the same temporal inversion (s^{-1}). They differ only in what that inversion is applied to:

- **Frequency** ($1/s$) — the temporal inversion s^{-1} read directly against the axiom. Time alone, inverted.
- **Velocity** (m/s) — the same temporal inversion s^{-1} applied to length m . Time inverted, carried on space.

When the two are divided, their matching temporal depths meet:

$$\frac{\text{Velocity}}{\text{Frequency}} = \frac{m}{s} \times \frac{s}{1} = m \cdot s^0.$$

The shared s^{-1} has met itself, reducing the net operational count of the time dimension to zero. The remaining s^0 is the axiom of 1. There is no remaining mechanic and no active count on the time line—it has simply returned to its un-operated start. What remains is pure length m at count $c = 0$. The inversion pair has mechanically resolved a count-1 operation back down to the base floor.

20 The Complete Operational Unit Lattice (Counts 0 to 5)

21 The 4D Inverse Operation Boundaries: 720 Sphere and 512 Cube

At the closure count ($c = 5$), the system fully maps out the spatial boundaries of reality through two co-dependent, 4D inverse operation structures. These boundaries govern how rational space (the cube) and irrational movement (the sphere) mirror one another across the center. Both, when read as inverse operations across that center, reduce to the same single Axiom of 1.

21.1 The 720 Angular 4D Inverse Operation Sphere

The uncut vector 720° is machined directly from the active lattice coordinates ($5 \times 9 \times 16$) at count 0. It represents a complete, multi-dimensional spherical rotation mapped in 4D space. The system operates symmetrically from a true center where inverse mechanics read both negative and positive orientations concurrently ($-$ and $+$).

Read as a dual-hemisphere inverse operation across that center, the two halves recover unity exactly:

$$\frac{720^\circ - 360^\circ}{360^\circ} = \frac{360^\circ}{360^\circ} = 1.$$

This is the $360/360 = 1$ identity: the spherical boundary, divided against its own hemisphere, returns to the Axiom of 1. The uncut 720° vector is the count-0 frame; each operational step down the lattice then partitions it into clean, rational angular landings (from 360° at count 1 down to the 22.5° duo-hexadecant at count 5). The same 720° frame bounds the irrational sphere boundary of 724.077 at the terminal depth $d = 9$.

21.2 The 512 4D Inverse Operation Cubic Square

In perfect structural opposition to the spherical rotation sits the 512 rational square boundary. This is the rigid, discrete spatial container—the cubic square of the lattice. Like the sphere,

c	d	Reading	Physical units	Mechanic
0	0	1 (rational)	meter (m), second (s), gram (g)	The axiom of 1. Un-operated base state. The floor where every unit sits and every net-zero operation naturally lands.
1	1	$2\sqrt{2}$ (irrational)	velocity (m/s), frequency ($1/s$), area (m^2)	First deviation. Single-unit operations. Velocity and Frequency form the temporal-inversion mirror; divided, they return to count 0.
2	2	8 (rational)	acceleration (m/s^2), volume (m^3), momentum ($g \cdot m/s$), density (g/m^3)	The rational accord. Compounding operations stabilise the irrational depth back into clean integers.
3	3	$8\sqrt{2}$ (irrational)	energy ($g \cdot m^2/s^2$), force ($g \cdot m/s^2$), pressure ($g/(m \cdot s^2)$)	The energy balance point. The m^2 and s^2 are mirrored inversion counts; they balance each other without either reaching zero, anchoring the reading to the invariant weight (g) line.
4	4	128 (rational)	power ($g \cdot m^2/s^3$)	The power vector. One further time inversion (s^{-3}) flips the depth back to a rational state.
5	8	512 (rational)	gravitational constant ($G = m^3/(g \cdot s^2)$)	The rational closure boundary. The dimensions saturate and lock into the 512 square boundary. The edge of the lattice.

Table 18: The complete operational unit lattice from the axiom of 1 to closure. Each count records the net number of physical unit operations. A spent operation—one whose net exponent on a dimension reaches zero—drops that dimension back to $c = 0$, the axiom. A mirrored operation, where two dimensions carry equal and opposite counts without either reaching zero, holds its count and anchors to the invariant line.

read as an inverse operation across the central floor it returns to the axiom:

$$\frac{-256}{256} = -1 = 1.$$

This is the axiom read from across itself ($-1 = 1$): the count-1 dual (± 256), divided against itself, returns to the single unit. The uncut 512 vector is the count-0 container; its signed span runs from -256 to $+256$ about the center, the full boundary measuring 512. It is perfectly rational because it locks cleanly at the even master depth $d = 8$, where the domains (32) times the unit ($\sqrt{2}^8 = 16$) yield exactly 512.

21.3 The Geometric Recovery

These two matrices are not separate entities; they are the inverse faces of the same single physical unit. As recorded in Table 3, the large machining ratio of the sphere over the square ($\frac{720}{512} = 1.40625$), when multiplied by the small spherical lift ($\frac{724.077}{720}$), recovers the master mechanic exactly:

$$\frac{720}{512} \times \frac{724.077}{720} = \frac{724.077}{512} = \sqrt{2}.$$

The cube provides the rational framework, the sphere provides the continuous rotational sweep, and the $0.7071 = 1/\sqrt{2}$ inverse ladder ensures that the Axiom of 1 is maintained perfectly at every active count through simple Pythagorean recovery. Both boundaries, reduced across their own center, return to the same axiom: $360/360 = 1$ for the sphere, $-256/256 = -1 = 1$ for the cube.

22 The Inverse Mechanical Count of the Axiom of 1

The constant $0.7071 = 1/\sqrt{2}$ is not an arbitrary geometric ratio; it is the inverse mechanic reading its own reflection directly off the master table at every active count. As detailed in Table 10, when the irrational unit of any count ($\sqrt{2}^d$) is divided by that count's domains (2^c), it structurally returns $1/\sqrt{2}$ at every step. Because the center reads inverse states concurrently ($-$ and $+$), this value represents the exact balance point of the unit's expansion.

When these coordinate tracks interact, they perform a pure inverse mechanical count that recovers the Axiom of 1 without relying on external geometric assumptions:

- The value $0.7071 = 1/\sqrt{2}$ is the universal inverse mechanic active across all counts.
- The 45° alignment is the structural piece size at count $c = 4$ on the 720 angular vector.
- The 90° orthogonality is the structural piece size at count $c = 3$ on the 720 angular vector.

When two matching inverse tracks ($1/\sqrt{2}$) collide under the master mechanic, their squaring forces a return to the axiom:

$$\left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2 = \frac{1}{2} + \frac{1}{2} = 1.$$

Each squared track resolves to $\frac{1}{2}$ —the count-1 dual halving—and the two halves sum to the single unit. This is the machine recovering its own floor. It is a simultaneous reading of three master-table coordinates—count 3, count 4, and the inverse ladder—colliding mechanically to prove that no matter how far the lattice deviates into irrational depths, the net operation always resolves directly back to the single physical unit at the Axiom of 1.

23 Section 5 — Planck Quantities on the Count

Standard physics builds Planck units by combining $\hbar$, G , and c as composite algebraic bricks to cancel human dimensions. Under the Axiom of 1, these quantities are not constructed by arithmetic; they are the specific physical readings manifested when the master mechanic halves the foundational unit to a specific geometric partition layer. Count c is strictly the halving mechanic—the structural depth—and nothing else.

A phenomenon is seated at a count by the highest geometric dimension required to contain it: $c = 0$ the un-halved scalar floor, $c = 1$ the linear vector, $c = 2$ the plane, $c = 3$ the volumetric grid, $c = 4$ the tensor, $c = 5$ the saturated closure. The standard formulas are not the cause of the seat; they are algebraic readouts that cross-lock with the reading at that halving depth. Where a formula's tally runs negative, that negative is the scar of forcing a low-dimensional phenomenon through a compounded denominator—not the source of the seat.

24 Section 6 — The Unified Count Matrix

Each count is a geometric dimension fixed by the slicing of the 720° sphere and the 512 cube, and the parity of that count's depth fixes the nature of the space: even depths read rational

Planck quantity	Standard formula	Halving c	Geometric partition alignment
Planck Time t_P	$\sqrt{\hbar G/c^5}$	$c = 0$	Un-halved floor: pure temporal duration anchor.
Planck Mass m_P	$\sqrt{\hbar c/G}$	$c = 0$	Un-halved floor: pure scalar substance anchor.
Planck Length ℓ_P	$\sqrt{\hbar G/c^3}$	$c = 1$	1st halving: linear spatial extension (360°).
Planck Force F_P	c^4/G	$c = 1$	1st halving: linear dynamic translation (360°).
Planck Pressure	$c^7/(\hbar G^2)$	$c = 1$	1st halving: directional push; highest independent dimension is linear.
Planck Charge q_P	$\sqrt{\hbar c}$	$c = 2$	2nd halving: planar/orthogonal constraint (180°).
Planck Momentum p_P	$m_P c$	$c = 2$	2nd halving: planar translational momentum (180°).
Planck Area A_P	ℓ_P^2	$c = 3$	3rd halving: the octant grid (90°).
Planck Power P_P	c^5/G	$c = 3$	3rd halving: volumetric action rate (90°).
Planck Energy E_P	$m_P c^2$	$c = 4$	4th halving: hexadecant stress-energy tensor (45°).
Planck Volume V_P	ℓ_P^3	$c = 5$	5th halving: total discrete container capacity (22.5°).
Planck Density ρ_P	$c^5/(\hbar G^2)$	$c = 5$	5th halving: terminal vacuum packing limit (22.5°).

Table 19: The twelve Planck quantities seated by geometric partitioning, not by algebraic tally. The halving count c is the literal structural dimension of the machine; the standard expressions are downstream readouts that land on these pre-existing coordinates. Seats are assigned by dimensional character: Charge and Momentum read as planar ($c = 2$) constraints, and Energy ($c = 4$) is the tensor seat while its rate Power ($c = 3$) reads at the volumetric grid.

(static containers), odd depths read irrational (dynamic vectors and boundaries carrying $\sqrt{2}$). Units land in their columns because their geometric behaviour matches the coordinate at the halving depth.

25 Section 7 — The Structural Speed of Light Squared ($c^2 = 1024$)

In standard physics, the speed of light squared ($c^2 \approx 8.988 \times 10^{16} \text{ m}^2/\text{s}^2$) carries human-constructed dimensions. Under the Axiom of 1, this constant is the downstream physical shadow of a pure lattice structural limit. The master mechanic defines the terminal capacity of the system at the fifth halving count ($c = 5$), where the discrete square boundary reaches 32 domains (2^5).

The absolute rational square of this closure boundary produces the dimensionless lattice constant:

$$\text{Lattice Constant} = 32^2 = 1024.$$

The number 1024 is not numerically equal to the dimensioned value of c^2 ; rather, it represents the invariant structural coordinate that mirrors the behaviour of c^2 within a closed geometry. It is the total spatial capacity of the machine before closure locks the grid.

Count c (dimension)	Even d rational	Odd d irrational ($\times\sqrt{2}$)
$c = 0$ (scalar)	$d = 0$ (1): Planck Time / Mass — the floor	—
$c = 1$ (1D)	—	$d = 1$ ($2\sqrt{2}$): Planck Length / Force / Pressure, Quantum Spin, RMS
$c = 2$ (2D)	$d = 2$ (8): Planck Charge / Momentum	$d = 3$ ($8\sqrt{2}$): Compton / Particle Wavelength
$c = 3$ (3D)	$d = 4$ (32): Planck Area / Power	$d = 5$ ($32\sqrt{2}$): Escape Velocity / Kinetic Boundary
$c = 4$ (tensor)	$d = 6$ (128): Planck Energy	$d = 7$ ($128\sqrt{2}$): Relativistic Momentum
$c = 5$ (closure)	$d = 8$ (512): Planck Volume / Density / G	$d = 9$ ($512\sqrt{2}$): Singularity Horizon / Time Flux

Table 20: The unified count matrix under geometric seating. The dimensional label sits on the count, covering both parities at that depth. Count $c = 1$ has no rational partner because a single linear slice of the Axiom cannot possess a squared area; Planck Pressure sits at $c = 1, d = 1$ as a 1D directional vector with Length and Force. The terminal boundary at $c = 5, d = 8$ locks the maximum spatial volume, the maximum density limit, and the Gravitational Constant (G) as the final rational walls of the lattice.

26 Section 8 — Mass-Energy Equivalence ($E = mc^2$) on the Lattice

When the mass-energy equivalence formula ($E = mc^2$) is mapped to the Master Operational Unit Lattice, it reveals a state of total domain saturation rather than an algebraic collapse back to the floor.

We map the components strictly by their lattice coordinates:

- **Mass (m):** sits at the un-halved anchor floor ($c = 0$), yielding a scalar reading of 1.
- **Speed of Light Squared (c^2):** the total spatial capacity count of 1024 (Section 7).
- **The Partition Divisor:** the fifth halving count splits space into 32 discrete domains (2^5).

The mechanic runs the interaction of the un-halved anchor against the total spatial capacity across the terminal partition:

$$E = 1 \times \frac{1024}{32} = 32.$$

The output is 32. This is the exact rational reading of the master table at Count $c = 3$, depth $d = 4$ ($\sqrt{2^8} \rightarrow 8 \times 4 = 32$). The lattice shows that energy is not a separate substance: it is the total spatial capacity of the closure limit (1024) compressed and read directly through the domains of the fifth halving, landing on a clean rational lattice value.

27 Section 9 — The Identity Rotation: $M^2/F^2 = 1$

At the absolute terminal boundary of the lattice ($c = 5$), the discrete container and the continuous vector achieve total mechanical equilibrium. Here, the total mass-energy tracking capacity (M) and the maximum structural force constraint (F) both saturate at the domain value of 32.

When these dual vectors operate against each other as a squared ratio, they perform an identity rotation:

$$\frac{M^2}{F^2} = \frac{32^2}{32^2} = \frac{1024}{1024} = 1.$$

While this resolves to 1 by construction, its mechanical significance is that the maximum boundaries of mass and force grind through every geometric depth of the lattice only to return to absolute identity. The boundary locks itself from within, establishing the stable floor of the macro-universe.

28 Section 10 — The Fine-Structure Constant Cross-Lock

Because the inverse fine-structure constant (α^{-1}) is entirely dimensionless, its comparison to the lattice is a legitimate, non-tuned structural verification. The integer core 137 is read two independent ways on the 512 rational vector, both plus the terminal depth coordinate ($d = 9$):

$$\text{Count } c=2 \text{ (quadrant): } 128 + 9 = 137, \quad \text{Count } c=4 \text{ (hexadecant): } (4 \times 32) + 9 = 137.$$

The quadrant piece 128 and the four hexadecant pieces $4 \times 32 = 128$ are the same rational reading approached from two counts; both plus the terminal depth 9 give the identical core 137.

The fractional remainder is the same terminal depth 9 read against the count-1 dual ($256 = 2 \times 128$):

$$\frac{9}{256} = 0.03515625.$$

Summing the core and the inverse lift yields the complete machine value:

$$\alpha_{\text{machine}}^{-1} = 137 + 0.03515625 = 137.03515625.$$

Compared to the standard physics value ($\alpha^{-1} \approx 137.035999$), the two agree to five significant figures (137.03). The absolute difference is 0.00084275, a structural agreement of 0.0006%.

29 Section 11 — Machine π and the 137 Internal Consistency Proof

Like α^{-1} , the circular constant π is fundamentally dimensionless, making it an un-tuned geometric property of the lattice. The machine reads π at Count $c = 3$ using three pure coordinates: the count itself (3), the count-3 octant piece size on the 512 vector (64), and the terminal depth (9):

$$\pi_{\text{machine}} = \frac{(3 \times 64) + 9}{64} = \frac{201}{64} = 3.140625.$$

This produces a match to standard π (3.14159265) with a structural difference of just 0.031%.

The defining proof of this section is not its proximity to nature, but its absolute internal consistency. When the octant denominator (64) is subtracted from the machine- π numerator (201), the lattice recovers the fine-structure core exactly:

$$201 - 64 = 137.$$

Two entirely independent geometric expressions—one defining the circular sweep of the machine (π), the other defining the electromagnetic core (α^{-1})—share the same numerical key. This cross-lock is an internal structural property of the lattice, not a fit to experimental data.

Coordinate	Value	Master-table source
Thermal dual base	$2\sqrt{2}$	$c = 1, d = 1$ irrational
Count-4 rational seat	128	$c = 4, d = 6$ rational
Count itself (numerator add)	+4	$c = 4$
Terminal depth (denominator add)	+9	$c = 5, d = 9$
Numerator	$128 + 4 = 132$	seat plus count
Denominator	$128 + 9 = 137$	seat plus terminal depth
Reading	$264\sqrt{2}/137$	—
Decimal	2.72519 K	—
Measured CMB	2.72548 K	Planck-era measurement
Difference	0.00029 K (0.011%)	count reading vs measured

Table 21: The CMB temperature on the count. The count-1 thermal dual $2\sqrt{2}$ reduced through the count-4 absorption structure $(128+4)/(128+9)$ gives $264\sqrt{2}/137 \approx 2.72519$ K; the measured CMB is 2.72548 K, a difference of 0.011%.

30 Section 12 — CMB Temperature on the Count

The cosmic microwave background temperature reads at count 4. The count-1 thermal dual is reduced through the count-4 absorption structure.

31 Section 13 — 0.7071 in Standard Physics

Standard physics finds $0.7071 = 1/\sqrt{2}$ across many domains. All appearances are the same single reading: the irrational unit divided by the count’s domains at every active count.

Standard phenomenon	Standard value	Count reading
Hadamard superposition $ \psi\rangle = 0.7071 0\rangle + 0.7071 1\rangle$	0.7071	inverse mechanic at every count
Probability $ 0.7071 ^2 = 0.5$	50%	count-1 dual halving
$\cos 45^\circ = \sin 45^\circ$	0.7071	inverse mechanic / count-4 angle (720/16)
RMS factor (AC effective value)	0.7071	inverse mechanic
Audio half-power threshold (−3 dB)	0.7071	inverse mechanic
Side of 45°-45°-90° unit triangle	0.7071	Pythagorean recovery of axiom

Table 22: The 0.7071 in standard physics. Every appearance of $1/\sqrt{2}$ in standard physics is the same single reading on the count: irrational unit divided by domains at every active count. Quantum superposition, AC RMS, 45° trigonometry, and audio half-power thresholds are all the master mechanic’s inverse, viewed through different domain-specific vocabularies.

32 Section 14 — G on the Count

Gravity reads at count 5, depth 8, the rational closure of the master table. The count reads G as the inverse from 1 of the count-5 rational closure, with the operational unit ratio $\text{m}^3/(\text{g} \cdot \text{s}^2)$ supplying the unit lens.

Coordinate	Construction	Reading
Count-5 rational closure	$32 \times 16 = 512$	$c = 5, d = 8$
Inverse from 1	$1/512$	0.001953125
Ops	$\text{m}^3/(\text{g} \cdot \text{s}^2)$	physical unit operation ratio
G on the count	$(1/512) \cdot \text{m}^3/(\text{g} \cdot \text{s}^2)$	the count reading with the ops
Standard G_{SI}	$6.67430 \times 10^{-11} \text{ m}^3/(\text{kg} \cdot \text{s}^2)$	same 1/512 reading in human units

Table 23: G on the count. The count reads G as the inverse from 1 of the count-5 rational closure 512, multiplied by the operational unit ratio $\text{m}^3/(\text{g} \cdot \text{s}^2)$. The standard SI value 6.67430×10^{-11} is the same 1/512 reading translated through the human kg-m-s unit system.

33 Section 15 — Schwarzschild and Kerr on the Count

The Schwarzschild and Kerr radii use the count-5 frequency squared ($c^2 = 1024$) and the count-1 dual (2). Substituting these into the standard formulas gives the count's reading directly.

Standard piece	Standard value	Count reading	Source
2 in $2GM/c^2$	2	$2^1 = 2$	$c = 1$ domains
M	external (kg, $M_{\odot}$)	axiom at gauge	the axiom itself
c^2	8.988×10^{16}	1024	$c = 5$ frequency squared
$R_s = 2GM/c^2$	standard form	$GM/512$	rational closure denominator
$R_+ = GM/c^2$ (Kerr)	half radius	$GM/1024$	no count-1 dual
R_s/R_+	2	count-1 dual	the 2 that disappears at max spin

Table 24: Schwarzschild and Kerr radii on the count. The Schwarzschild radius substitutes the count's $c^2 = 1024$ and reads as $R_s = GM/512$, with the count-5 rational closure in the denominator. Maximum-spin Kerr collapses the count-1 dual back to the axiom: only one of the two halves remains.

Constant	Standard value	Count reading	Difference / status
α^{-1}	137.035999	$137 + 9/256 = 137.03515625$	0.0006% (agrees to 5 sig figs)
π	3.14159265	$201/64 = 3.140625$	0.032%
T_{CMB}	2.72548 K	$264\sqrt{2}/137 = 2.72519$ K	0.011%
c^2	8.988×10^{16}	1024 ($c = 5 f^2$)	exact on the count
G	6.67430×10^{-11}	$(1/512) \cdot \text{m}^3/(\text{g} \cdot \text{s}^2)$	through gauge
720° spinor cover	720°	720° machined ($5 \times 9 \times 16$)	exact
$0.7071 = 1/\sqrt{2}$	0.7071	unit / domains at every c	exact
0.5 (probability)	0.5	count-1 dual halving	exact
$\hbar$	action, $\text{g} \cdot \text{m}^2/\text{s}$	$c = 3, d = 3$ irrational	operational seat
Planck Length ℓ_P	1.616×10^{-35} m	$c = 1, d = 1$ irrational ($2\sqrt{2}$)	dynamic wave-vector
Planck Mass m_P	2.176×10^{-8} kg	$c = 0, d = 0$ floor	axiom of weight at rest
Planck Time t_P	5.391×10^{-44} s	$c = 0, d = 0$ floor	axiom of time at rest
Planck Energy E_P	1.956×10^9 J	$c = 3, d = 4$ rational (32)	axiom as energy

Table 25: All standard constants side by side. Every constant standard physics treats as a separate quantity reads at a specific master-table coordinate. The difference column states the count reading against the standard value directly; constants with units carry the human observer’s gauge inside them.

34 Section 16 — All Standard Constants Side by Side

35 Section 17 — Conclusion

Conclusion — The Rosetta Stone Closed.

There is exactly one physical unit — the axiom of 1. Three base measurements (length, time, weight) are three views of that single axiom. Every standard derived unit is a count of physical unit operations against the axiom, with count c equal to the number of operations minus one. The floor rule holds the axiom invariant: zero and negative results collapse back to the axiom, never below.

Standard physics constants read at specific master-table coordinates:

- Fine-structure reading $137+9/256 = 137.03515625$ at $c = 2$ (body $128+9$) and $c = 4$ (body $4 \cdot 32 + 9$), inverse lift $9/256$ on the count-1 dual; standard $\alpha^{-1} = 137.035999$, agreeing to five significant figures (0.0006%).
- Machine $\pi = 201/64 = 3.140625$ at $c = 3$, with cross-lock $201 - 64 = 137$.
- CMB $T = 264\sqrt{2}/137 \approx 2.72519$ K at $c = 4$ (count-1 thermal through count-4 absorption).
- $c^2 = 1024 = 32^2$ at $c = 5$ frequency squared.
- $G = (1/512) \cdot \text{m}^3/(\text{g} \cdot \text{s}^2)$ at $c = 5, d = 8$ rational closure.
- $0.7071 = 1/\sqrt{2}$ at every count (unit / domains).
- Twelve Planck quantities placed at $c = 0$ through $c = 5$ via unit arithmetic with the floor rule.
- Every count carries a rational (even d) and an irrational (odd d) reading; static physical properties on the rational side, dynamic waves and limits on the irrational side.

The framework closes at $c = 5$. Standard physics never exceeds this count because count 5 is the lattice closure itself. The mathematics is the reading; the single physical unit is the thing read.